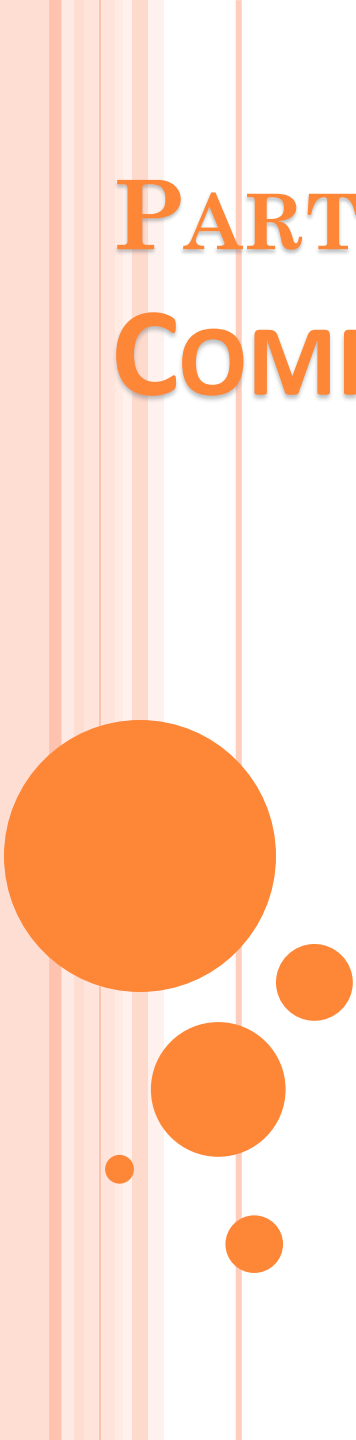




PART2

COMPUTER ARITHMETIC

**Computer Organization and
Assembly Language**

Computer Science Department

**National University of Computer and Emerging
Sciences Islamabad**

MULTIPLICATION

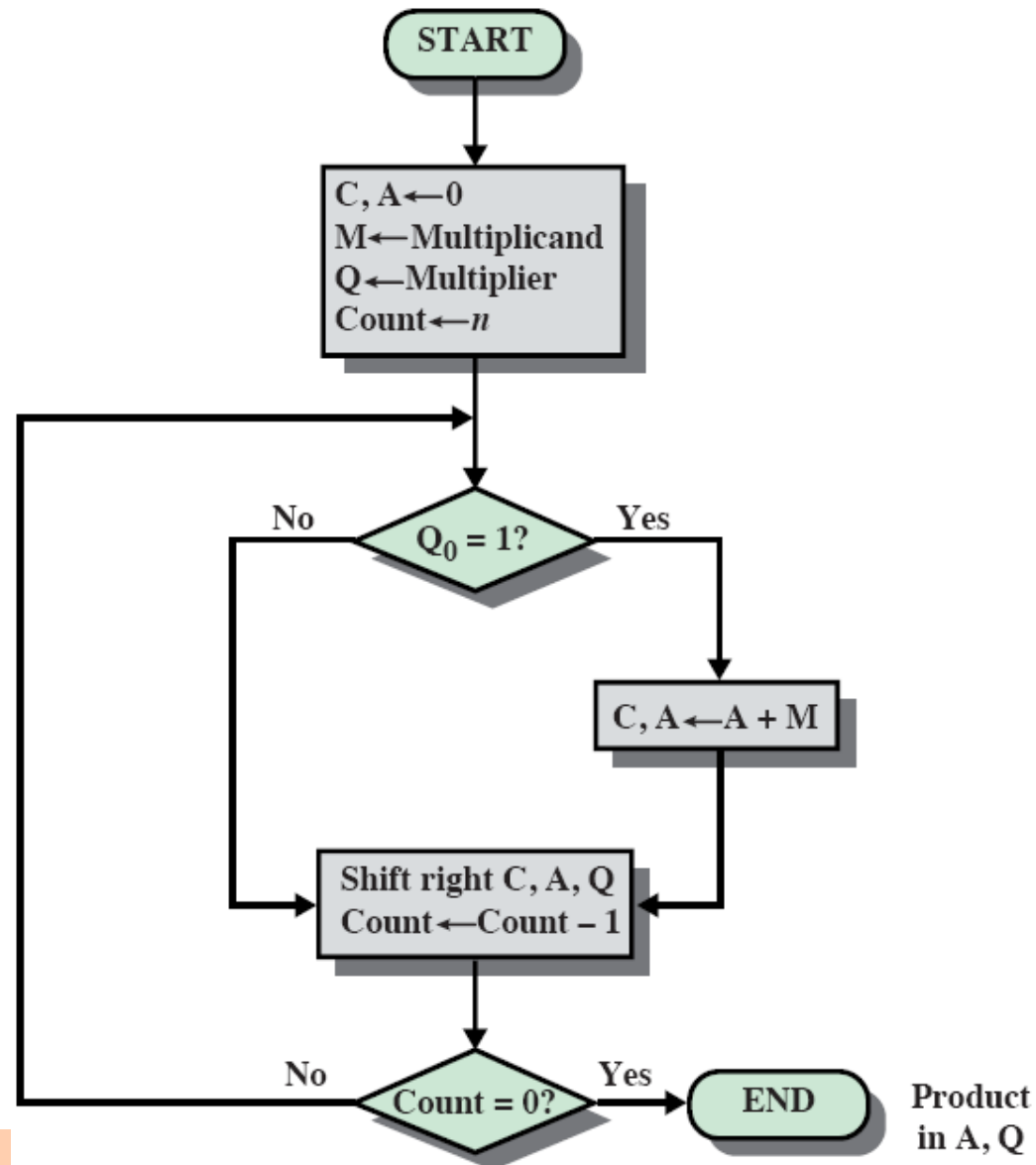
- Complex
- Work out **partial product** for each digit
- Take care with place value (column)
- Add partial products

MULTIPLICATION EXAMPLE

- 1011 Multiplicand (decimal 11)
- x 1101 Multiplier (decimal 13)
- 1011 Partial products
- 0 000x Note: if multiplier bit is 1,
copy multiplicand
- 10 11xx otherwise zero
- 101 1xxx
- 1000 1111 Product (11x13=143 decimal)

- Note: length of product is the sum of lengths of multiplicand and multiplier
- 4-bits x 4-bits => 8-bits answer

UNSIGNED BINARY MULTIPLICATION



EXECUTION OF EXAMPLE

C	A	Q	M	Initial Values	
0	0000	1101	1011		
0	1011	1101	1011	Add	} First Cycle
0	0101	1110	1011	Shift	
0	0010	1111	1011	Shift	} Second Cycle
0	1101	1111	1011	Add	
0	0110	1111	1011	Shift	} Third Cycle
1	0001	1111	1011	Add	
0	1000	1111	1011	Shift	} Fourth Cycle

* Add: $A = A + M$

Shift: Logical Shift Right

11001 Multiplicand M

11101 Multiplier Q

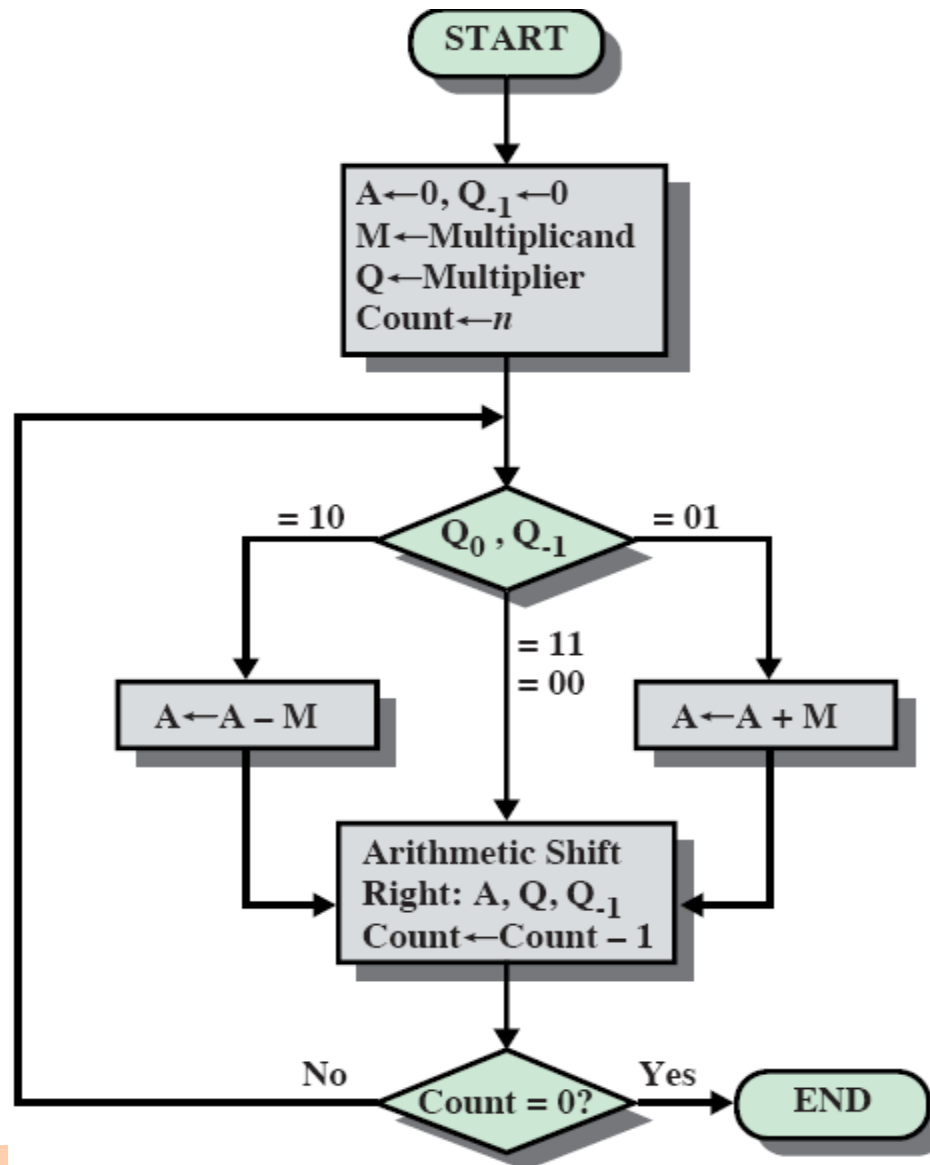
UNSIGNED MULTIPLICATION

Operation	C	Acc	Multiplier Q	M (Multiplicand)
Initialize	0	00000 +11001	11101	11001
Step1: Add M+A Shift R	0 0	11001 01100	11110	
Step2: Shift R	0 0	00110 +11001	01111	
Step3: M+A Shift R	0 0	11111 01111	01111 10111	
Step4: M+A Shift R	?	?	?	
Step5: M+A Shift R	0	10110	10101	

MULTIPLYING NEGATIVE NUMBERS

- This does not work!
- Solution 1
 - Convert to positive if required
 - Multiply as above
 - If signs were different, negate answer
- Solution 2
 - Booth's algorithm

BOOTH'S ALGORITHM



EXAMPLE OF BOOTH'S ALGORITHM

A	Q	Q ₋₁	M		
0000	0011	0	0111	Initial Values	
1001	0011	0	0111	A - M	First Cycle
1100	1001	1	0111	Shift	
1110	0100	1	0111	Shift	Second Cycle
0101	0100	1	0111	A + M	
0010	1010	0	0111	Shift	Third Cycle
0001	0101	0	0111	Shift	
					Fourth Cycle

* Shift: Arithmetic Shift Right i.e. sign bit is replicated

DIVISION

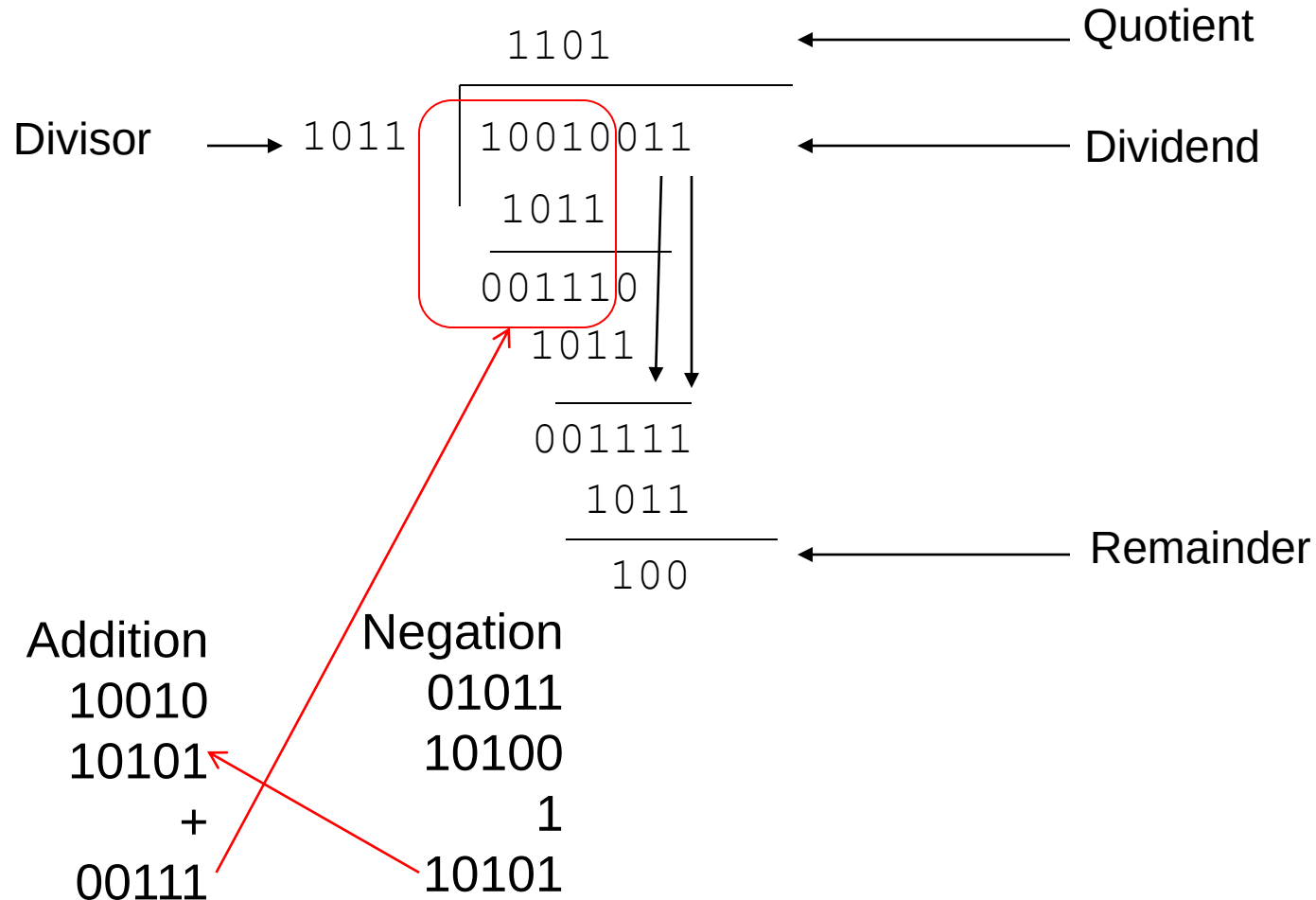
- More complex than multiplication
- Negative numbers are really bad!
- Based on long division

EXAMPLES

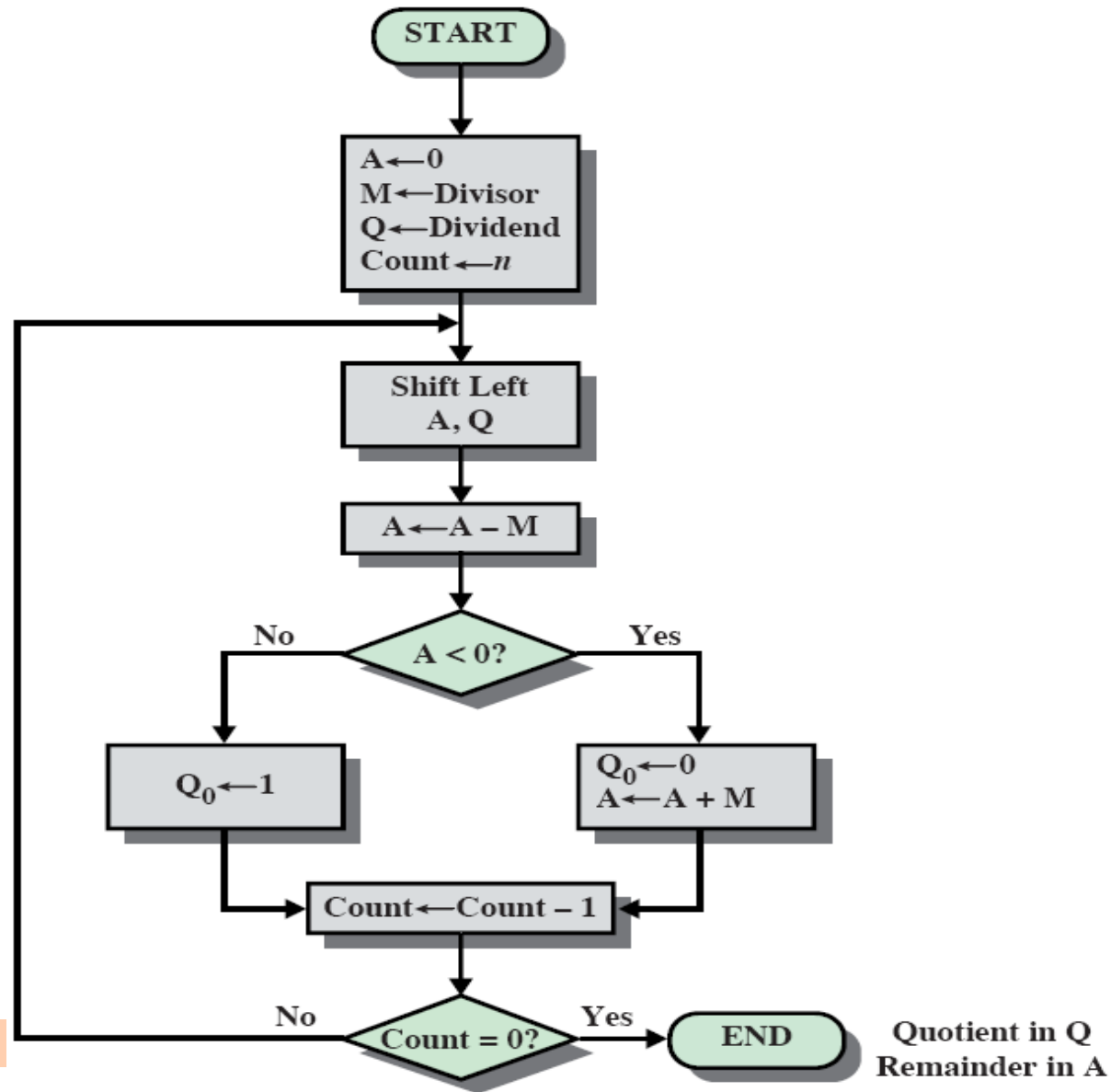
- $7(M) \times 3(Q) = 21$ (Signed Multiplication)
- $7(M) \times -3(Q) = -21$ (Signed Multiplication)
- $17(M) \times -11(Q)$ (Signed Multiplication)

DIVISION OF UNSIGNED BINARY INTEGERS

*** 147/11=> Q=13 ; R=04**



FLOWCHART FOR UNSIGNED BINARY DIVISION



UNSIGNED BINARY DIVISION

A	Q	
0000	0111	Initial value
0000	1110	Shift
<u>1101</u>		Use two's complement of 0011 for subtraction
1101		Subtract
0000	1110	Restore, set $Q_0 = 0$
0001	1100	Shift
<u>1101</u>		
1110		Subtract
0001	1100	Restore, set $Q_0 = 0$
0011	1000	Shift
<u>1101</u>		
0000	1001	Subtract, set $Q_0 = 1$
0001	0010	Shift
<u>1101</u>		
1110		Subtract
0001	0010	Restore, set $Q_0 = 0$

Questions & Answers